Planetary Orbits and Speed of Light from Pure Geometry
(No Newtonian G)

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Abstract

We derive a geometric speed scale and approximate orbital radii of the terrestrial planets from a compact geometric seed

$$S = \phi^6 \times 31,$$

where ϕ is the golden ratio.

The framework combines dimensionless geometric ratios, spatial projection operators, and simple scaling relations involving $\sqrt{3}$, factorial structures, and the decimal metric scale.

Using these constructions, approximate values are obtained for:

- the speed of light scale,
- Earth's orbital speed,
- the solar gravitational parameter,
- the orbital radii of Mercury, Venus, and Mars.

The resulting values numerically approach empirical astronomical data with sub-percent deviations. The relations are presented as exploratory geometric correlations rather than a replacement for established gravitational theory.

1 The Fundamental Geometric Seed

The golden ratio is defined as

$$\phi = \frac{1+\sqrt{5}}{2}.$$

The fundamental spiral step is introduced as

$$S = \phi^6 \times 31.$$

Numerically,

$$S \approx 556.272429 \text{ s}.$$

A geometric light-speed scale is then defined:

$$c_{\text{geom}} = (\phi^6 \times 31)^3 \times \sqrt{3} \times \left(1 + \frac{4}{720}\right).$$

Numerically,

$$c_{\text{geom}} \approx 2.99798395 \times 10^8 \text{ m/s}.$$

Reference SI value:

$$c_{\mathrm{SI}} = 2.99792458 \times 10^8 \text{ m/s}.$$

Relative deviation:

$$\approx 0.002\%.$$

2 Primary Spatial Metric

Define the baseline spatial metric:

$$L_{\mathrm{base}} = c_{\mathrm{geom}} \times S.$$

Numerically,

$$L_{\mathrm{base}} \approx 1.667668 \times 10^{11} \text{ m}.$$

This scale acts as the primary geometric reference length for the inner planetary system.

3 Earth Orbital Velocity from Geometry

Define the dimensionless geometric ratio

$$\Gamma = \frac{R_\odot \times R_{\mathrm{Moon}}}{R_\oplus^2},$$

where

$$R_\odot = 695700 \text{ km},$$

$$R_{\mathrm{Moon}} = 1737.4 \text{ km},$$

$$R_\oplus = 6371 \text{ km}.$$

Then

$$\Gamma \approx 29.7788.$$

Introduce a unit velocity scale

$$V_0 = 1 \text{ km/s}.$$

Earth's orbital speed becomes

$$v_\oplus = \Gamma \times V_0.$$

Numerically,

$$v_\oplus \approx 29.7788 \text{ km/s}.$$

Observed mean orbital velocity:

$$v_\oplus^{(\mathrm{obs})} \approx 29.7847 \text{ km/s}.$$

Relative deviation:

$$\approx 0.02\%.$$

4 Solar Gravitational Parameter

Using

$$v_{\oplus}^2 = \frac{GM_{\odot}}{r_{\mathrm{AU}}},$$

with

$$r_{\mathrm{AU}} = 149\,597\,870.7 \text{ km},$$

the geometric estimate becomes

$$GM_{\odot}^{(\mathrm{geom})} = r_{\mathrm{AU}} \times V_0^2 \times \Gamma^2.$$

Numerically,

$$GM_{\odot}^{(\mathrm{geom})} \approx 1.32660 \times 10^{20} \text{ m}^3\text{s}^{-2}.$$

Reference IAU/CODATA value:

$$GM_{\odot}^{(\mathrm{ref})} = 1.32712 \times 10^{20} \text{ m}^3\text{s}^{-2}.$$

Relative deviation:

$$\approx 0.04\%.$$

5 Terrestrial Planetary Orbital Radii

5.1 Mercury: Internal Compressed Node

Mercury is modeled as an internal compressed geometric node:

$$a_{\mathrm{Mercury}} = \frac{3\sqrt{3}}{5} L_{\mathrm{base}}.$$

Numerically,

$$a_{\mathrm{Mercury}} \approx 5.77702 \times 10^{10} \text{ m}.$$

Observed value:

$$5.79091 \times 10^{10} \text{ m}.$$

Relative deviation:

$$0.24\%.$$

5.2 Venus: Harmonic Mirror Node

Venus is modeled through a diagonal-factorial projection:

$$a_{\mathrm{Venus}} = 1 \text{ AU} \times \frac{5\sqrt{3}}{12}.$$

Equivalently,

$$a_{\mathrm{Venus}} = 1 \text{ AU} \times \frac{300\sqrt{3}}{720}.$$

Numerically,

$$a_{\text{Venus}} \approx 1.07963 \times 10^{11} \text{ m.}$$

Observed value:

$$1.08209 \times 10^{11} \text{ m.}$$

Relative deviation:

$$0.23\%.$$

5.3 Mars: External Expanded Node

Mars is modeled using an isometric volumetric expansion operator:

$$a_{\text{Mars}} = L_{\text{base}} \times \frac{\sqrt{3}}{\sqrt[3]{2}}.$$

Numerically,

$$a_{\text{Mars}} \approx 2.29259 \times 10^{11} \text{ m.}$$

Observed value:

$$2.27939 \times 10^{11} \text{ m.}$$

Relative deviation:

$$0.58\%.$$

6 Comparison Table

Planet	Geometric Value (m)	Observed Value (m)	Accuracy
Mercury	5.77702×10^{10}	5.79091×10^{10}	99.76%
Venus	1.07963×10^{11}	1.08209×10^{11}	99.77%
Mars	2.29259×10^{11}	2.27939×10^{11}	99.42%

Table 1: Geometric estimates compared with IAU values.

7 Emergent Planetary Skeleton

All three terrestrial orbital nodes emerge from the same baseline metric L_{base}:

$$a_{\text{Mercury}} = \frac{3\sqrt{3}}{5} L_{\text{base}},$$

$$a_{\text{Venus}} = \frac{360}{S} L_{\text{base}},$$

$$a_{\text{Mars}} = \frac{\sqrt{3}}{\sqrt[3]{2}} L_{\text{base}}.$$

where

$$S = \phi^6 \times 31.$$

This generates a compact geometric scaffold constructed from the restricted operator family

$$\{\phi, 31, \sqrt{3}, \sqrt[3]{2}, 5, 12, 360, 720\}.$$

Interpretationally:

- Mercury corresponds to geometric compression,
- Venus behaves as a harmonic mirror node,
- Mars emerges through volumetric expansion.

The appearance of

$$\frac{360}{S}$$

inside the Venus relation introduces an angular interpretation linking orbital scaling with rotational geometry.

8 Discussion

The framework demonstrates that several Solar System quantities can be approximated using a tightly constrained geometric structure involving:

$$\phi, \quad 31, \quad 720 = 6!, \quad \sqrt{3}, \quad \sqrt[3]{2}.$$

The model does not claim a dynamical replacement for celestial mechanics. Instead, it presents a family of persistent geometric correlations whose numerical convergence may warrant further mathematical investigation.

Residual deviations remain bounded below 1%, suggesting internal structural consistency within the proposed scaling relations.

9 Conclusion

A compact geometric framework based on the seed

$$S = \phi^6 \times 31$$

generates:

- a geometric light-speed scale,
- Earth's orbital speed,
- the solar gravitational parameter,
- approximate orbital radii for Mercury, Venus, and Mars.

The resulting relations reproduce empirical astronomical values with sub-percent deviations while using no free fitting parameters beyond the fixed geometric operators introduced in the framework.

Whether these correlations represent a deeper geometric structure or a high-order numerical coincidence remains an open question.

"I am almost joking. But the formulas work."

— Boris V. Kochenov, May 2026

Related Zenodo Records

1. Speed of Light, Gravity, and Mass from Geometry
   https://zenodo.org/records/20326444

2. mn/mp = 1 + 1/720
   https://zenodo.org/records/20146556

3. Geometric G From Earth's Geometry
   https://zenodo.org/records/20095666

4. The Integer 31 as a Discrete Geometric Node in Kochenov G
   https://zenodo.org/records/20071070

5. $\phi^{12}/\sqrt{5} = 144 + 1/720$
   https://zenodo.org/records/19539721